

Carbon Nanotube Transistors

KKKC6353 Nanoelectronic Devices · Lecture 4, Part 1 of 3

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Graphene: carbon one atom thick

- Carbon atoms in a **honeycomb**, each bonded to three neighbours
- One of the strongest materials known, and an excellent conductor of heat
- Its electrons move very fast, about **10^6 m/s**
- But it has **no band gap**, so a graphene transistor cannot be switched off

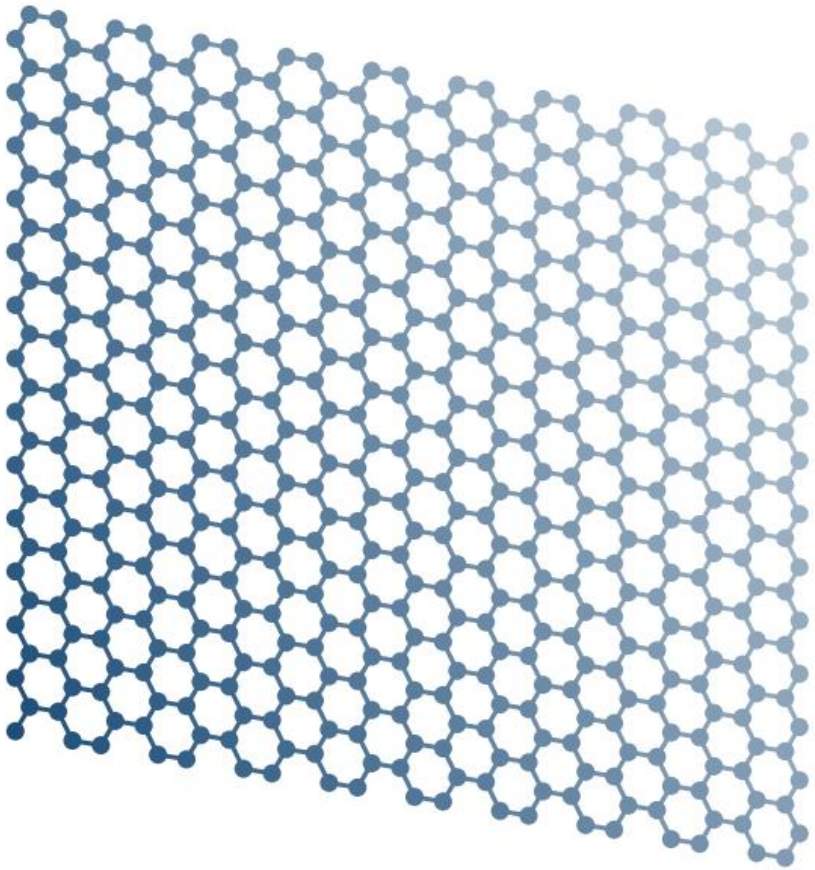
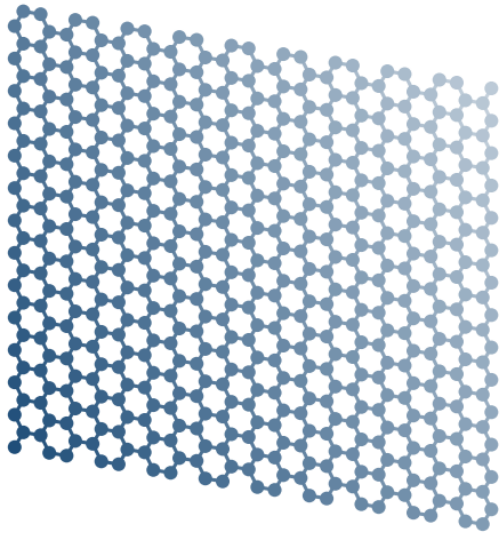


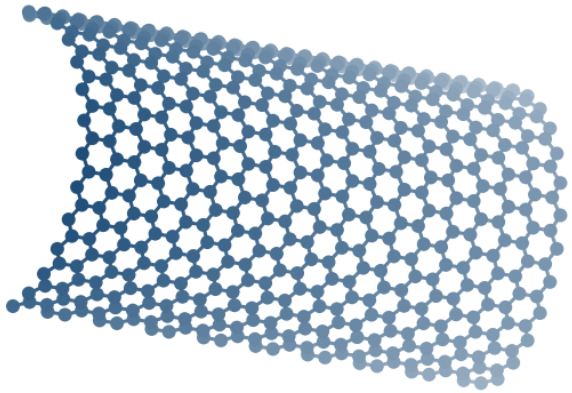
Figure made for this course.

Roll the sheet into a tube

1 Graphene: one sheet of carbon



2 Roll it up …



3 … into a seamless tube

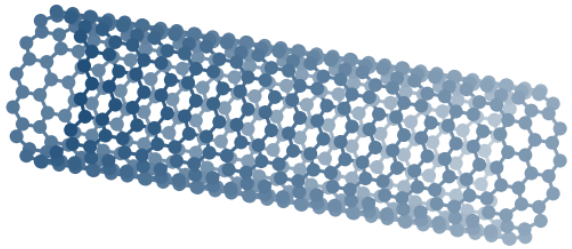
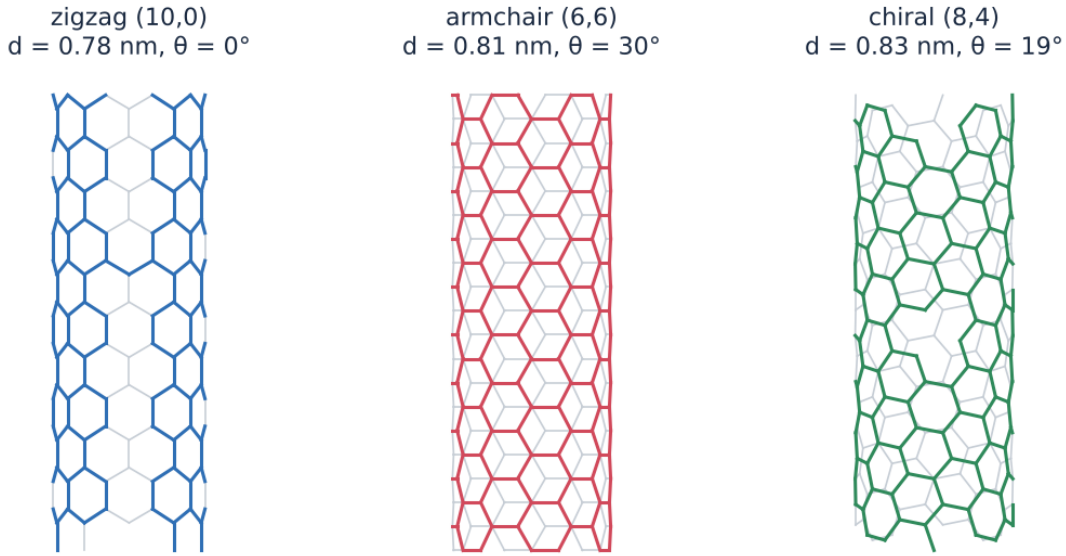


Figure made for this course: atom positions of a (14,0) tube, rolled step by step.

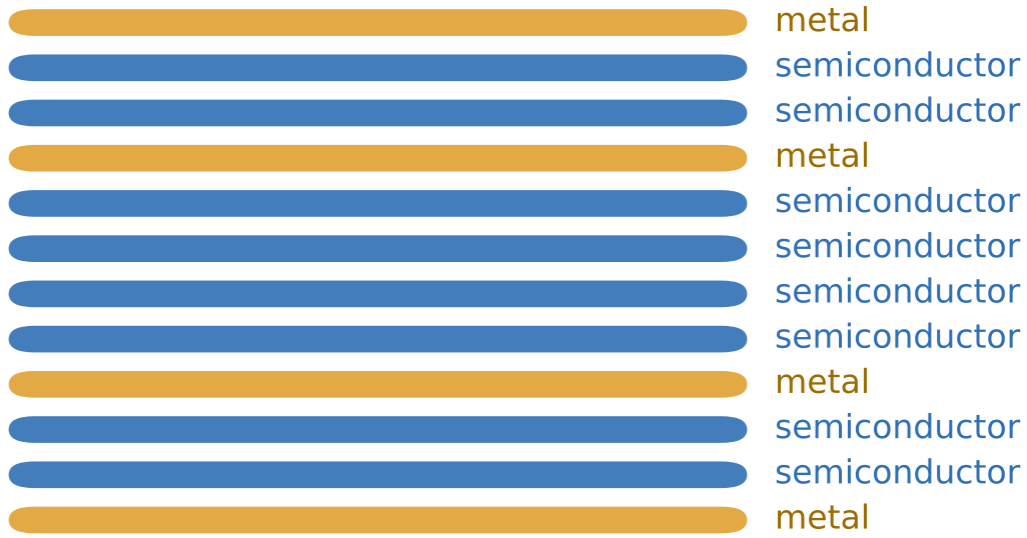
A **carbon nanotube** is a graphene sheet rolled into a seamless cylinder 1–2 nm across. It has no edges and no dangling bonds: an ideal surface to wrap a gate around.

How it is rolled decides metal or semiconductor



The three families of single-walled nanotube (side views, front wall coloured)

The three families of nanotube. Figure made for this course.



A typical growth: about one tube in three is metallic

Figure made for this course.

The angle of rolling decides the electrical type: **about one tube in three is a metal**. Growth gives a mixture, which is the main manufacturing headache.

Thinner tube, bigger gap

- The gap of a semiconducting tube is about **0.8 eV ÷ diameter in nm**
- A 1.5 nm tube: about **0.5 eV**, half of silicon's
- Wider tubes leak more; thinner ones are hard to contact
- Transistors use **1.2–1.8 nm** tubes

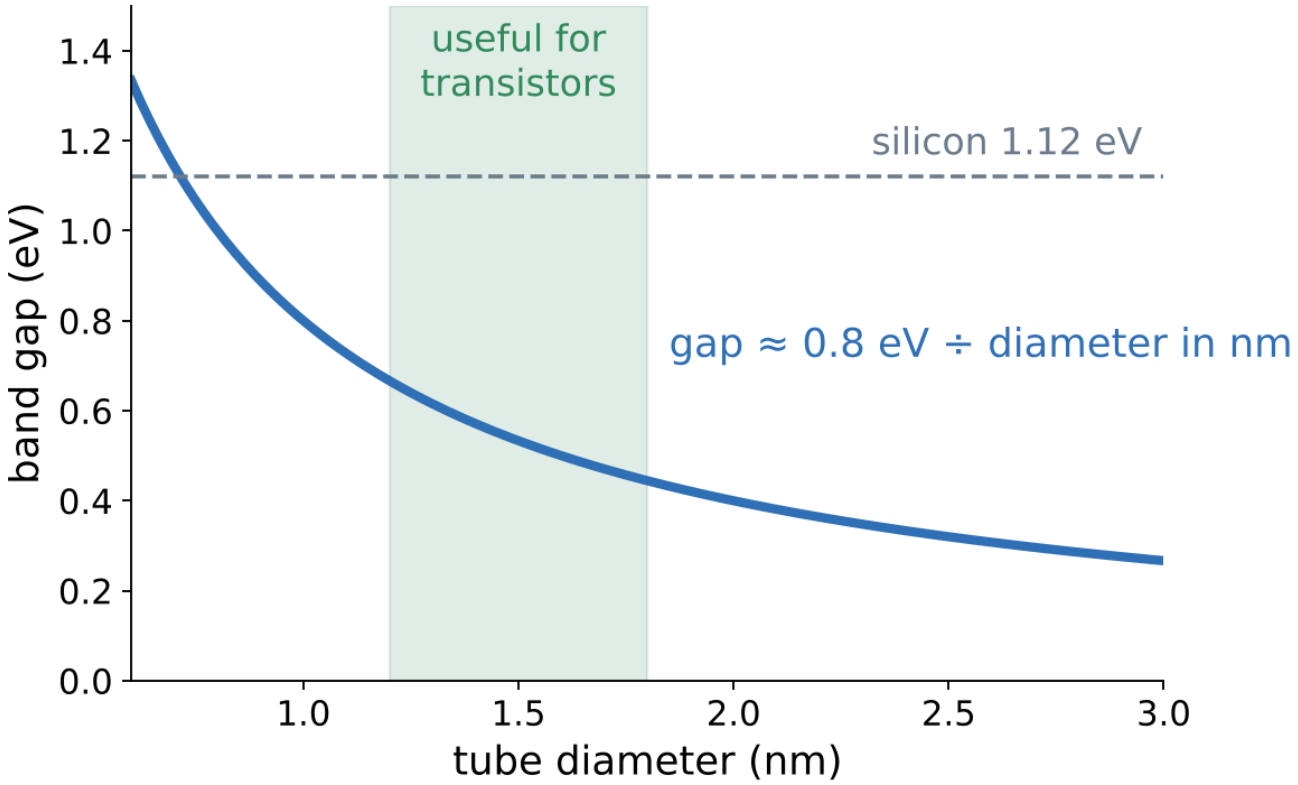


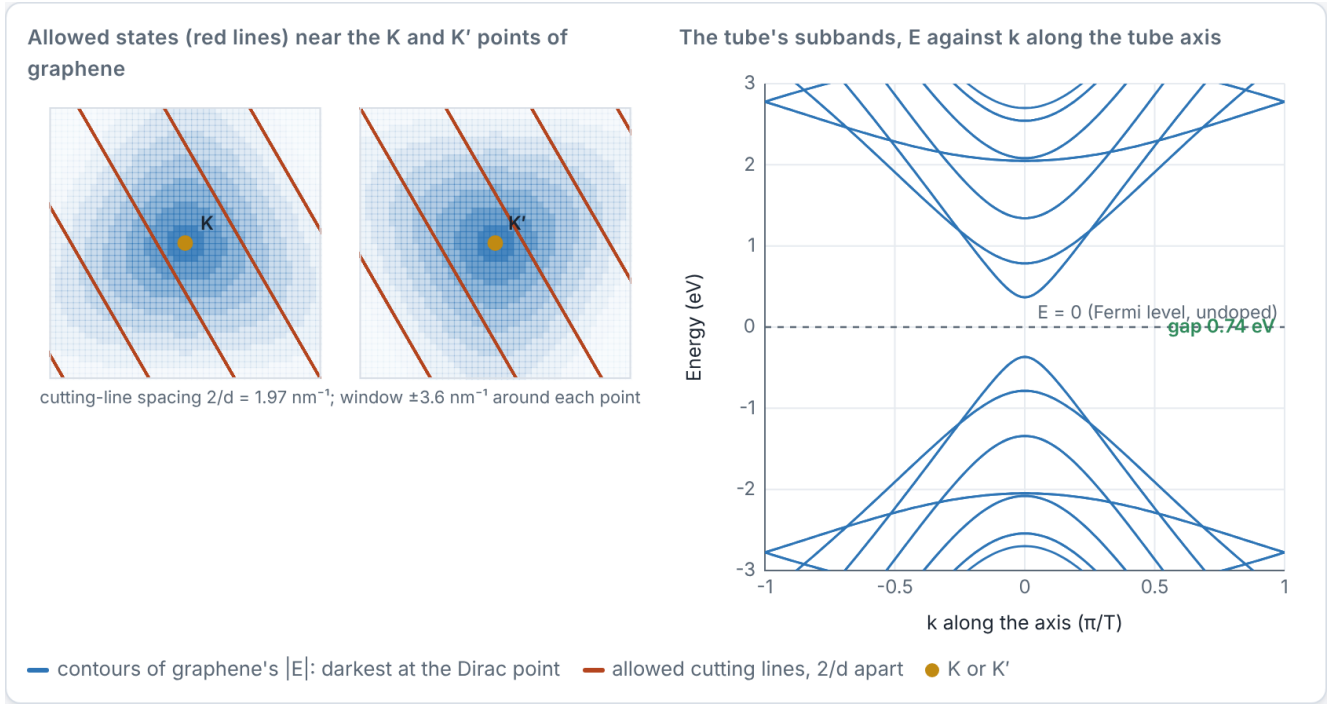
Figure made for this course; $E_g = 2a_{cc}\gamma_0/d$ with $\gamma_0 = 2.7$ eV.

Interactive animation 1

Nanotube band structure (optional)

faizverse.github.io/nanoelectronic-devices/L4/cnt-band-structure.html

- Compare (9,0), (10,0) and (11,0): which is the metal?
- Keep n − m fixed and widen the tube: what happens to the gap?
- For the curious: the band theory is in Document 1



1

The first nanotube transistors

One tube, two electrodes and a gate.

What a nanotube transistor looks like

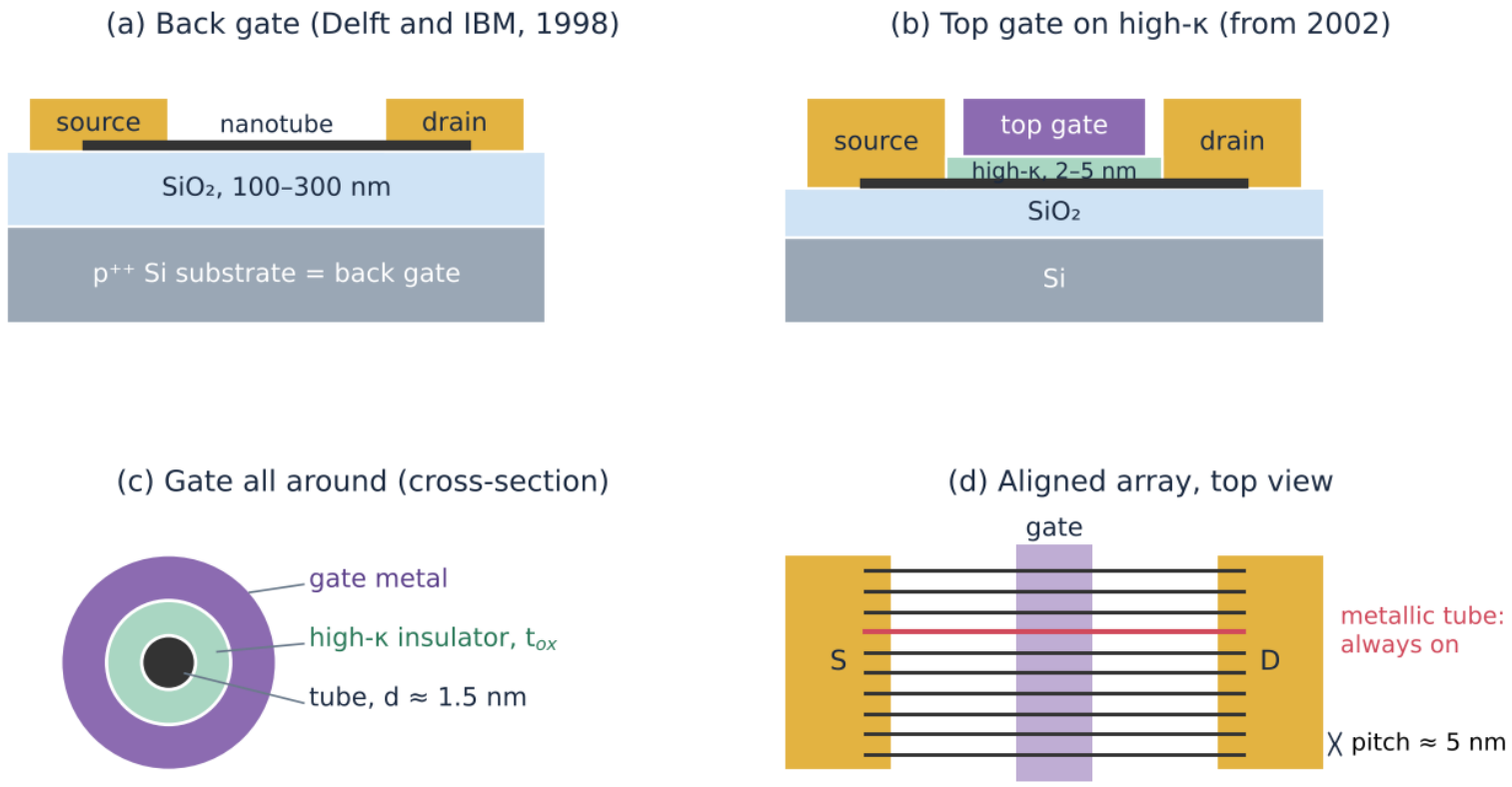
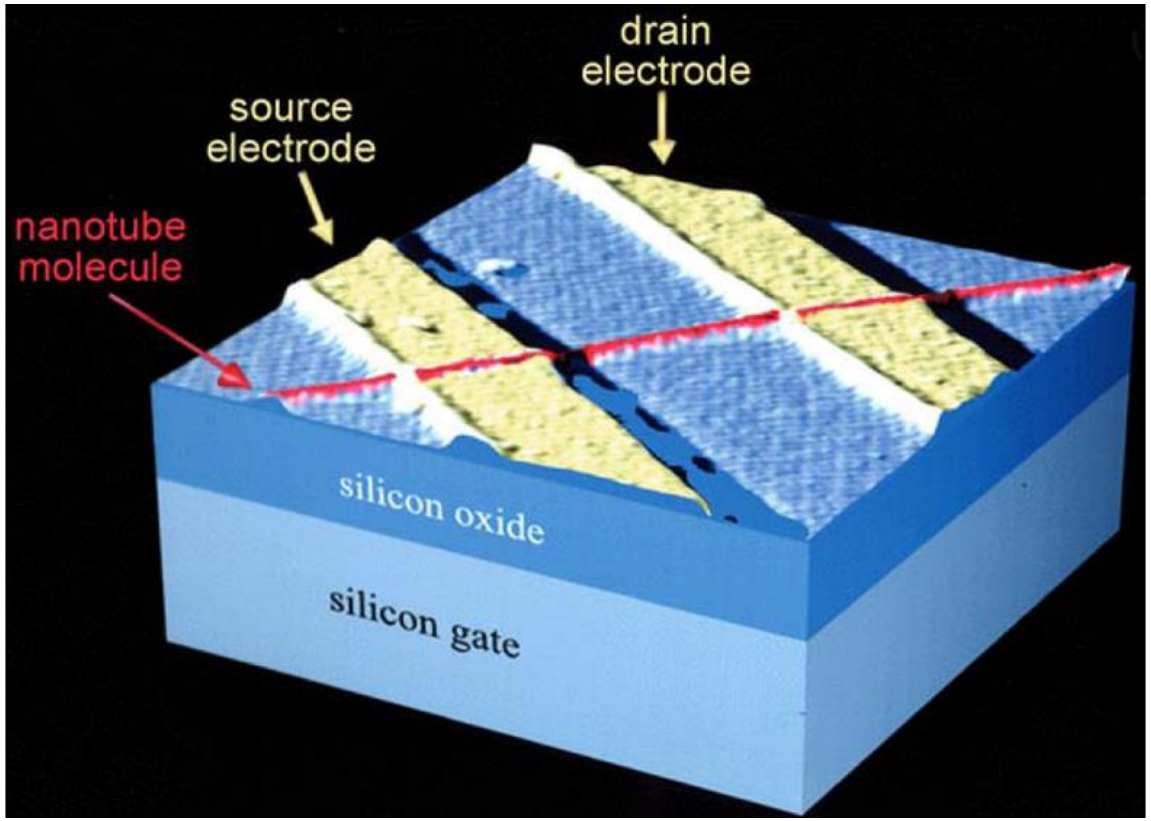
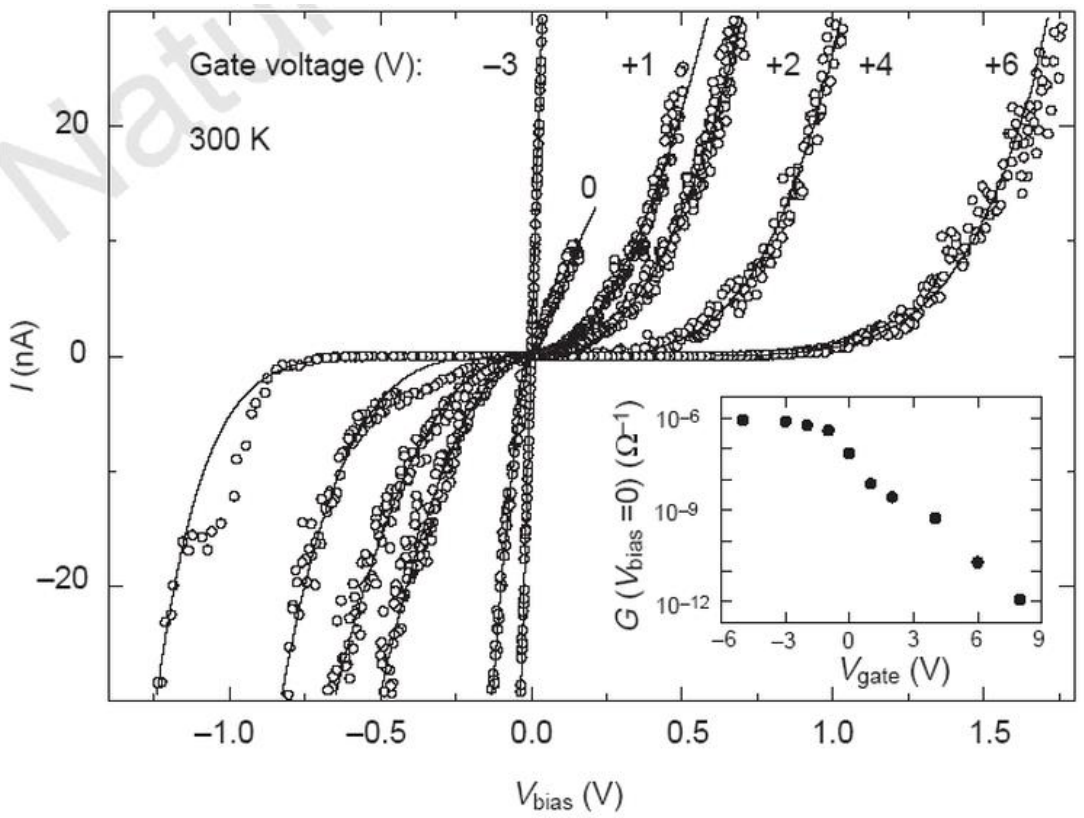


Figure made for this course.

From one tube on a back gate (1998) to a gate **all round** the tube, and then **arrays** of 100–200 tubes per µm, where one metallic tube cannot be switched off.

1998: the first nanotube transistors



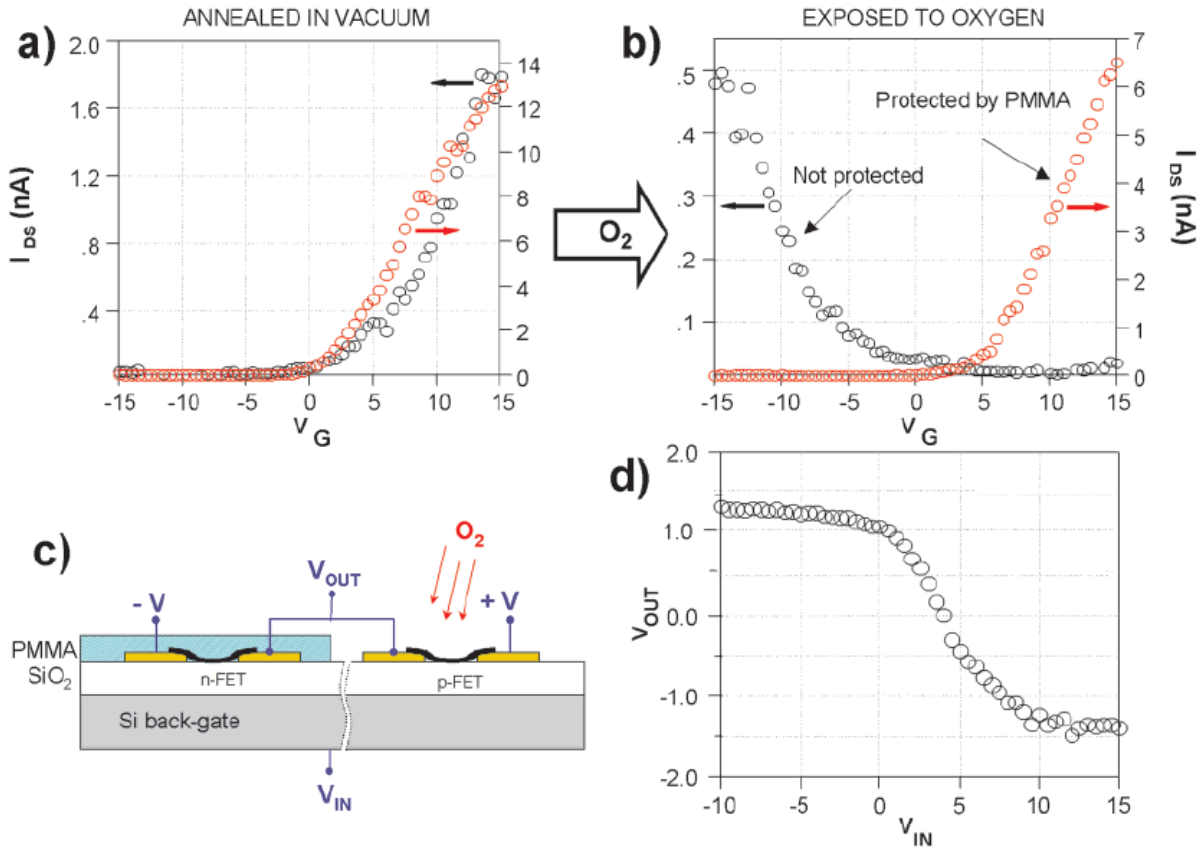


S. J. Tans, A. R. M. Verschueren and C. Dekker, Nature 393, 49 (1998); reproduced from T. Ragheb, Rice University ELEC 527 slides (2007).

One tube across two electrodes, the silicon wafer as the gate. It switched, but only as **p-type**, with megaohms of resistance and volts on the gate.

2001: n-type, and the first logic gate

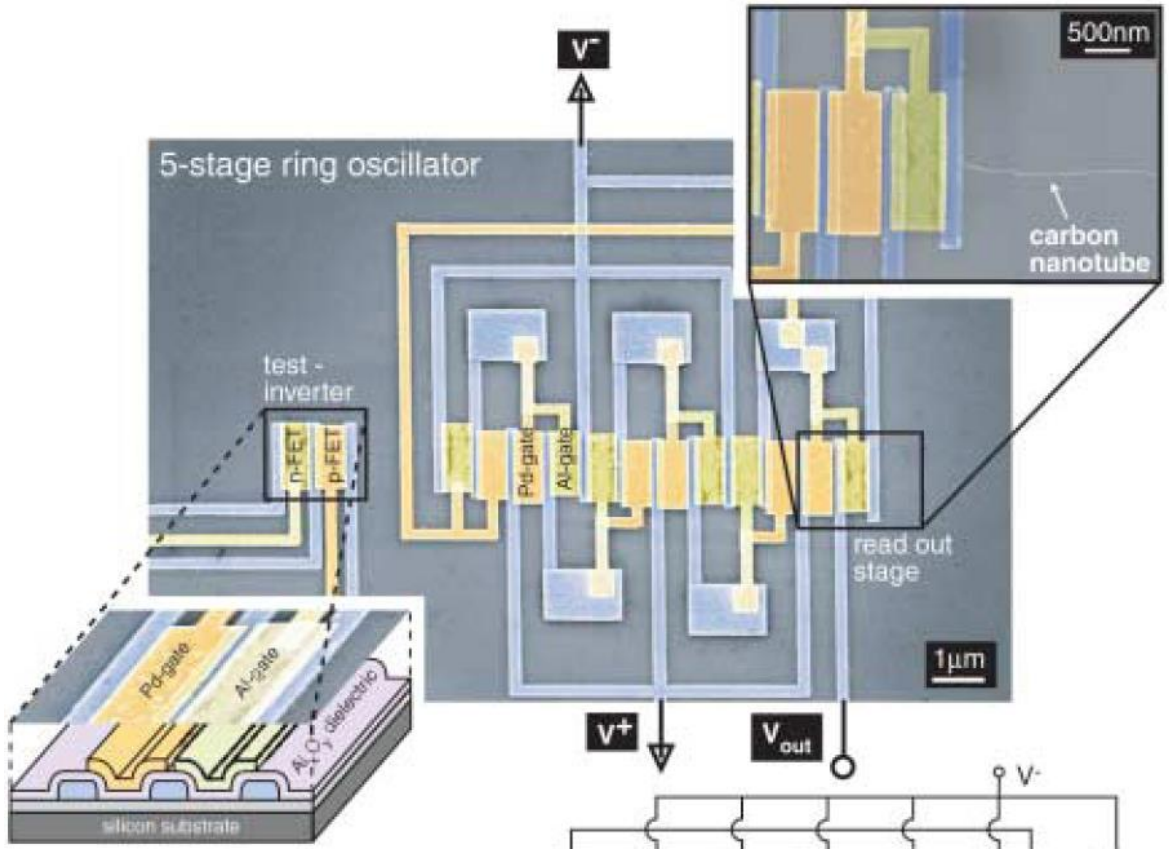
- Removing the oxygen made the tube **n-type**
- Half a tube p-type, half n-type: an **inverter**, the first logic gate
- Lesson: in a nanotube transistor the **contacts** decide how it behaves



V. Derycke et al., Nano Lett. 1, 453 (2001); reproduced from T. Ragheb, Rice University ELEC 527 slides (2007).

2006: a circuit along one tube

- Thin **high-κ** gates: switching at about 1 V
- Near-ideal contacts: **palladium** for holes (2003), **scandium** for electrons (2007)
- A five-stage **ring oscillator** built along one 18 µm tube



Z. Chen et al., Science 311, 1735 (2006); reproduced from T. Ragheb, Rice University ELEC 527 slides (2007).

2

Two ways to switch

Does the gate thin a barrier, or lower one?

Metal contacts: the gate thins a barrier

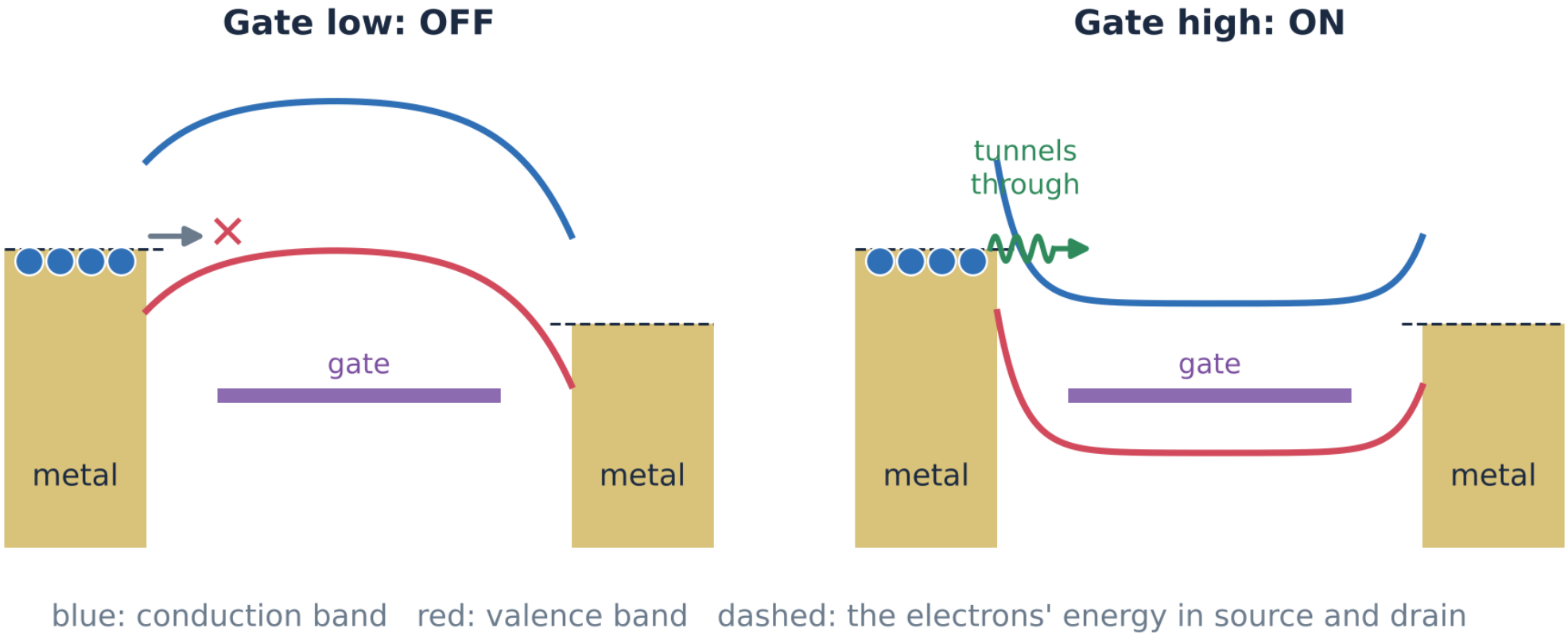


Figure made for this course.

Where metal meets tube there is a **barrier**. Gate low: it is thick and electrons are stopped. Gate high: the bands bend down, the barrier becomes **thin**, and electrons **tunnel** through.

Doped contacts: the gate lowers a hill

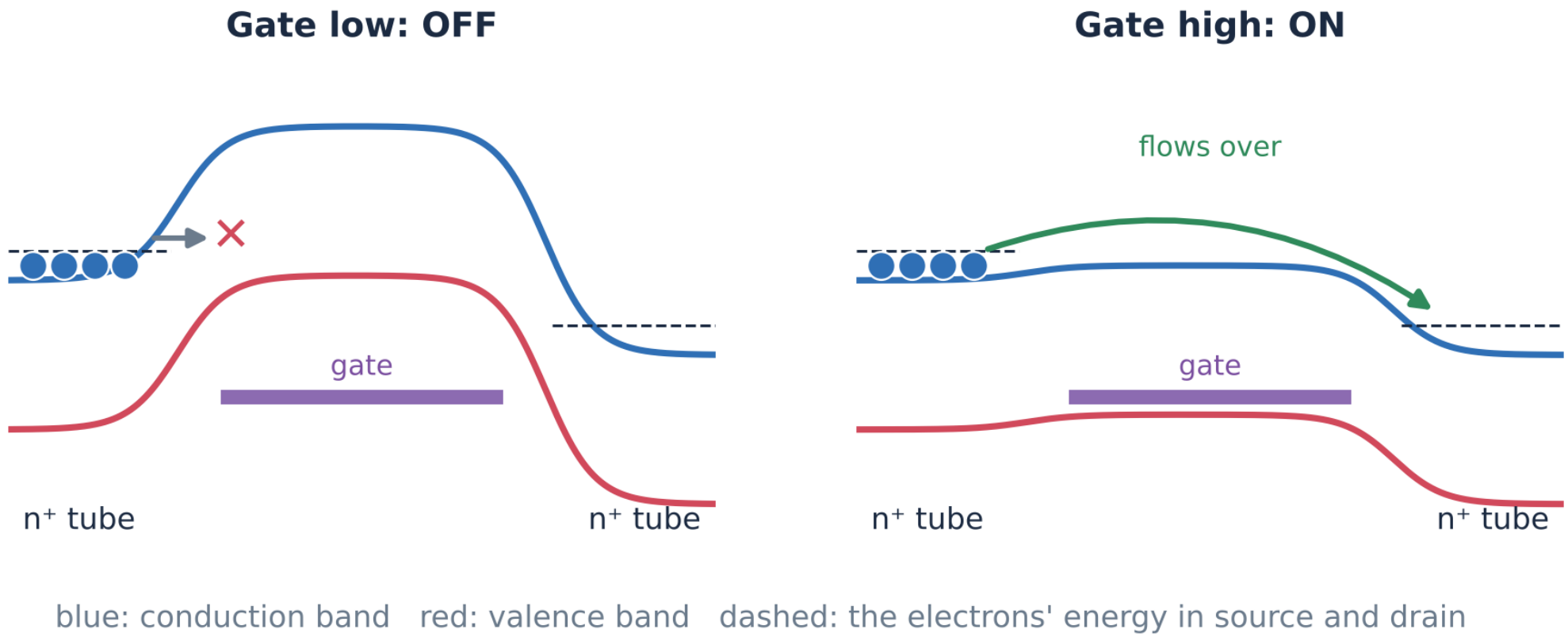


Figure made for this course.

With **doped** tube ends there is no contact barrier. The gate raises or lowers a **hill** in the channel and electrons flow **over** it, exactly as in a MOSFET.

Metal contacts let both carriers in

- Gate high: electrons tunnel in. Gate low: **holes** tunnel in. The curve is a **V**
- Wider tubes have smaller barriers, so they **leak more** when off
- A thinner gate oxide makes the barriers thinner: a **steeper** switch
- Doped contacts give one carrier only: the better transistor

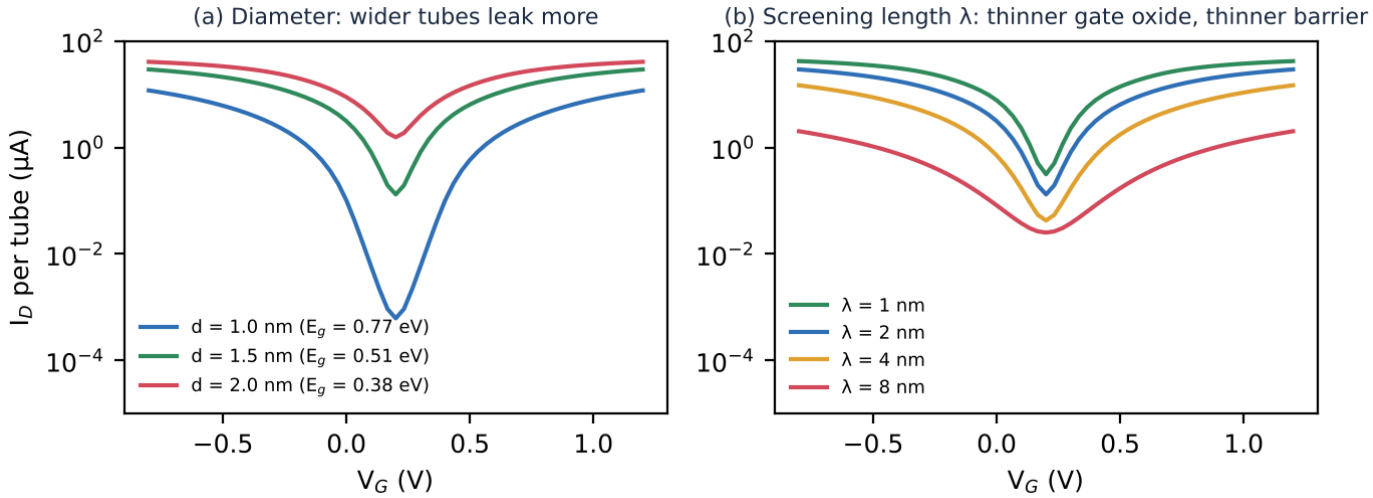


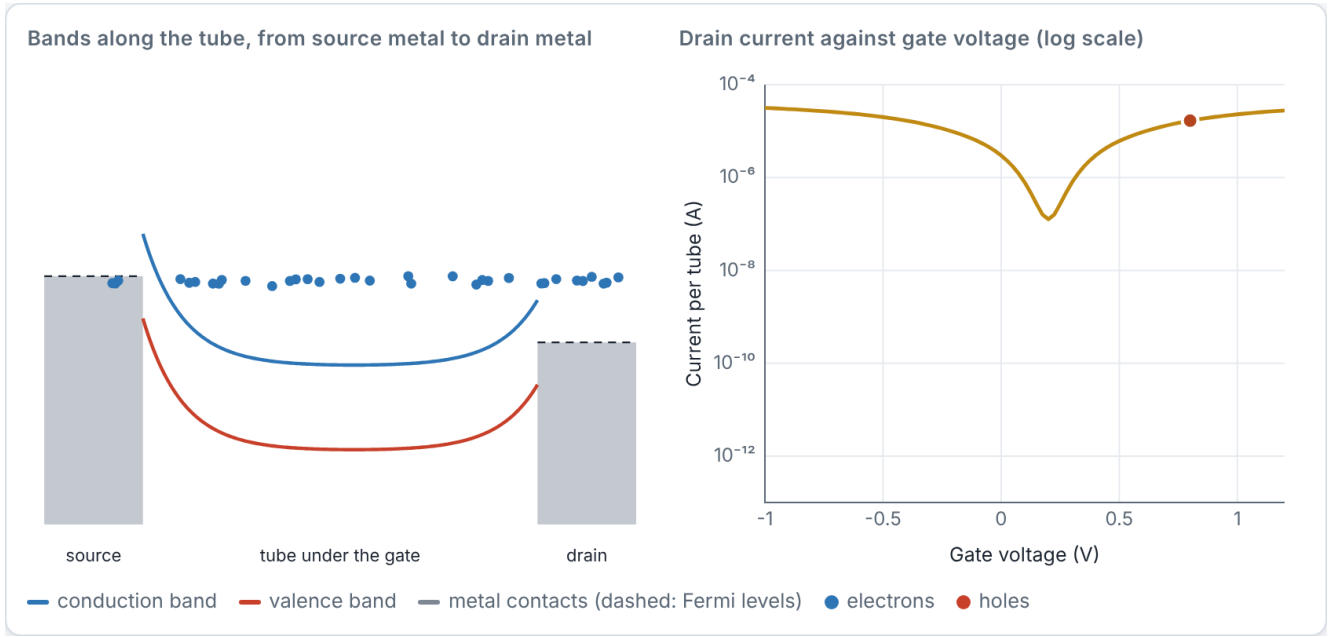
Figure made for this course.

Interactive animation 3

Schottky-barrier nanotube transistor

faizverse.github.io/nanoelectronic-devices/L4/sb-cntfet.html

- Where is the current smallest, and why?
- Widen the tube to 2 nm: what happens to the off-current?
- Make λ longer (thicker oxide): what happens to the slope?
- Switch to doped contacts and compare



3

The gate and the tube

Filling a tiny tube with electrons costs energy.

Two capacitors in series

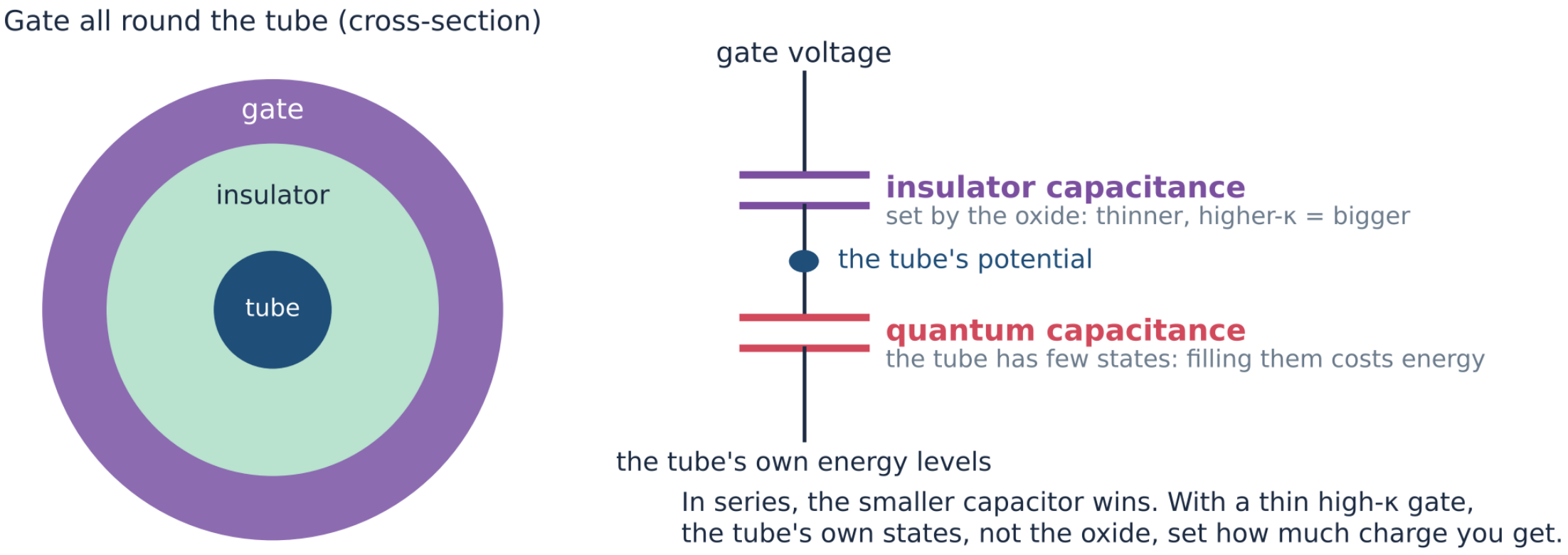


Figure made for this course.

The gate charges the tube through the **insulator**, but the tube has so **few states** that filling them takes energy too: a second, **quantum** capacitance in series.

What this means for the designer

Thick oxide

The insulator is the small capacitor: the gate has weak control, as in an old MOSFET.

Thin high-κ

The tube's states become the limit. The bands follow the gate closely: the steepest possible switch.

Diminishing returns

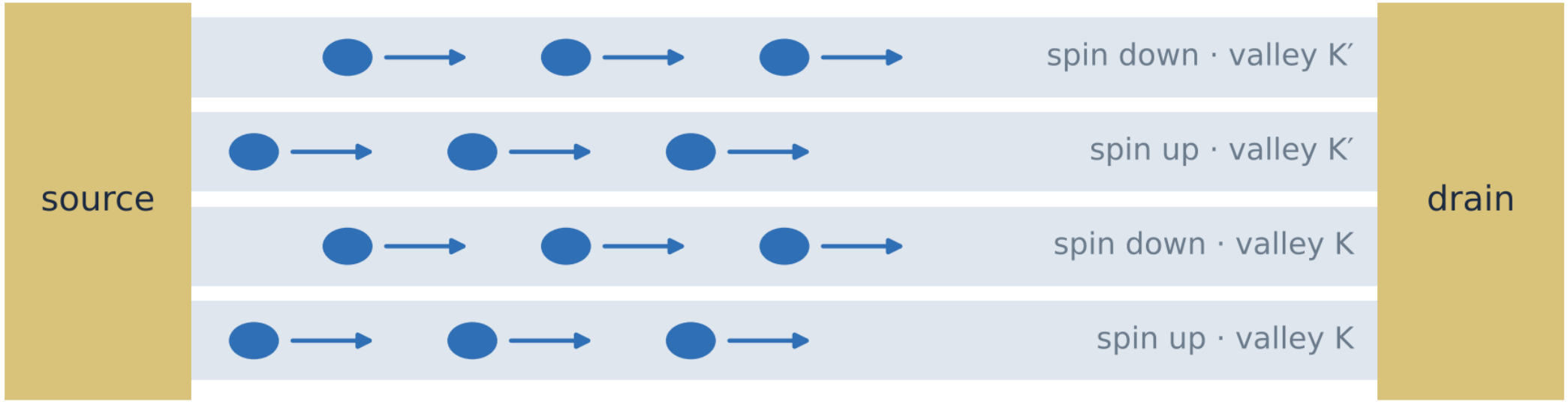
Below a few nm of HfO_2, a thinner oxide adds almost no charge. The tube itself sets the limit.

4

Ballistic transport

Four lanes, and no traffic lights.

Four lanes, no traffic lights



No collisions in a short tube: electrons fly straight through (ballistic). But each lane can carry only so much:

four lanes give a minimum resistance of $h/4e^2 = 6.45\ \text{k}\Omega$, paid at the contacts.

Figure made for this course.

In a tube shorter than about 1 µm electrons **do not collide**. Current is limited only by how many lanes there are, and the 6.45 kΩ is paid **at the contacts**.

What a perfect tube transistor would do

Sharp switch

About 60 mV per decade with a thin wrap-around gate: the thermal limit.

Early saturation

The current levels off at a small drain voltage.

About 40 µA

Per tube at 0.5 V; the quantum capacitance limits the charge.

Fast electrons

Injected two to four times faster than in silicon: this, not mobility, is what matters.

From one tube to mA per µm

40 µA

ideal current of one tube at 0.5 V

8 mA/µm

200 tubes per µm, if every tube were ideal

1–3 mA/µm

best measured arrays; silicon nanosheets give 1–2

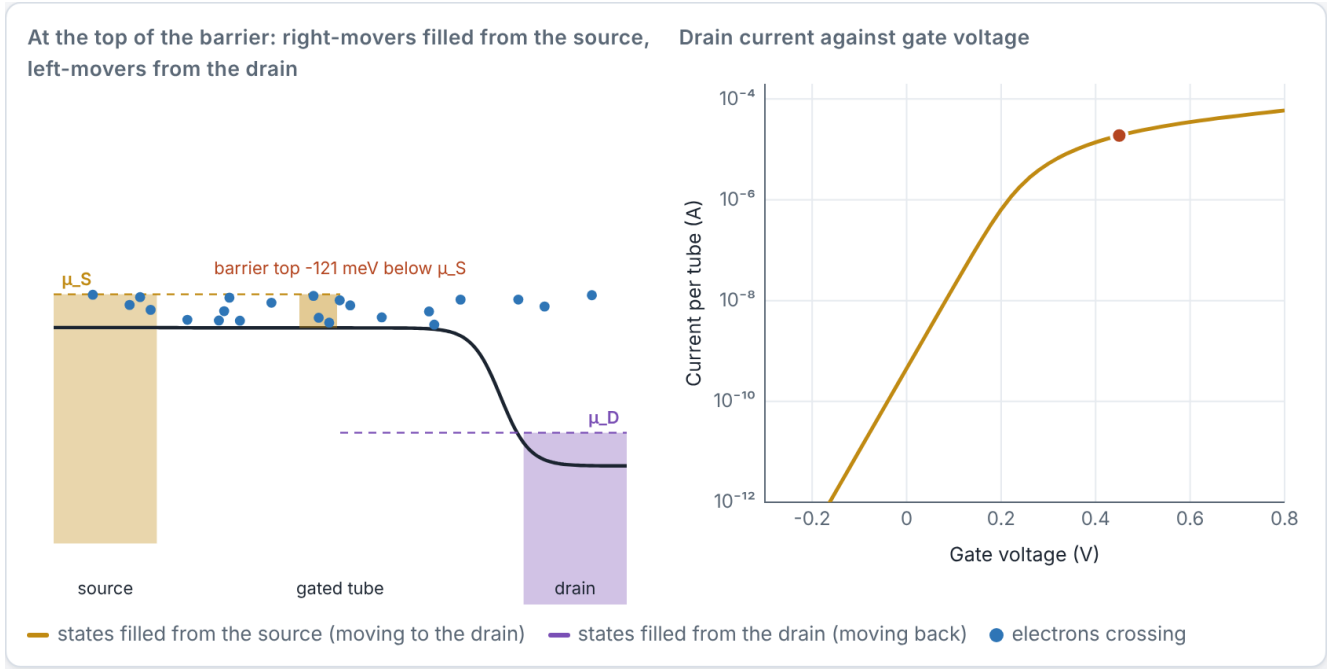
The gap between ideal and measured goes to **contacts**, uneven tube spacing, neighbouring tubes screening each other, and traps: engineering, not physics.

Interactive animation 2

The ballistic nanotube transistor

faizverse.github.io/nanoelectronic-devices/L4/cntfet-ballistic.html

- Sweep the gate: read the swing, then heat to 400 K
- SiO_2 → HfO_2, then thin it: when does the current stop rising?
- Lower the drain voltage: why does it saturate so early?
- Widen the tube: watch a second lane open

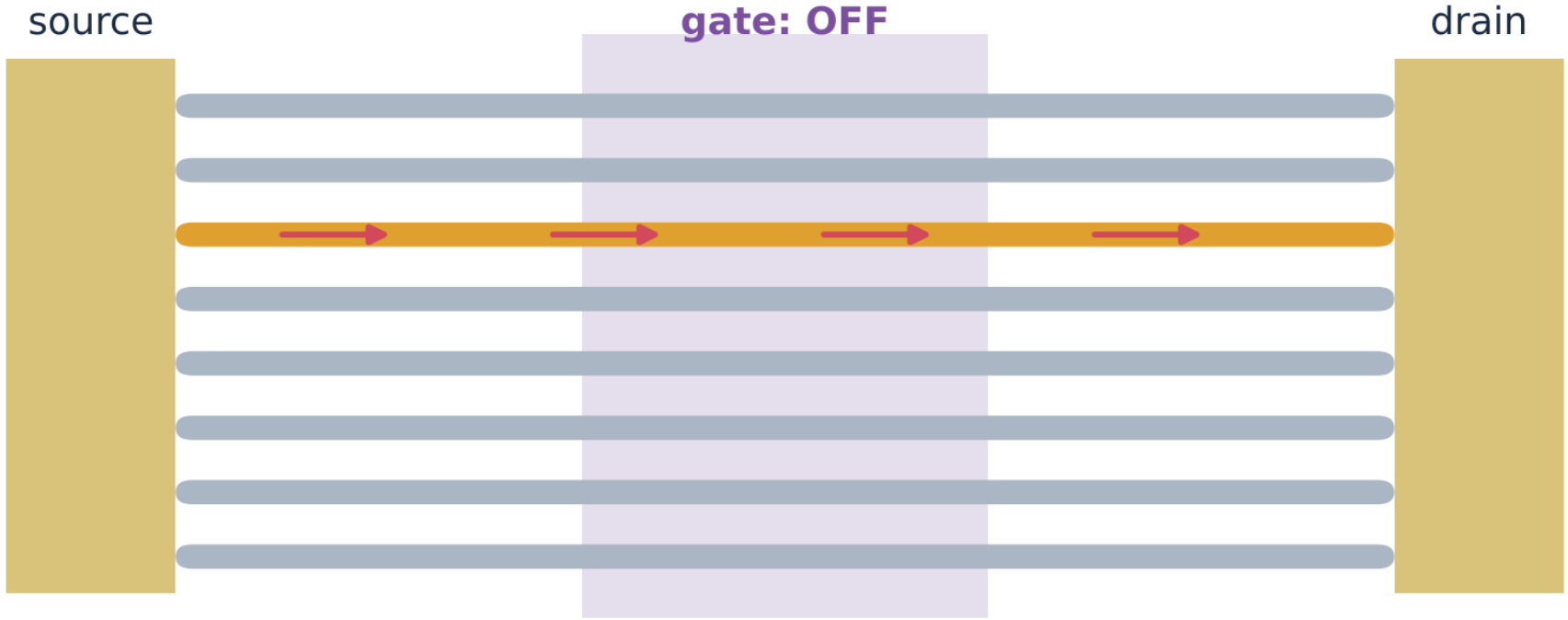


5

What stands in the way

The physics was settled by about 2012. Then came manufacturing.

One metallic tube spoils the transistor



Semiconducting tubes switch off (grey); one metallic tube (gold) keeps conducting: the transistor leaks.

Figure made for this course.

A metallic tube cannot be switched off. Large circuits need tubes that are **99.9999%** semiconducting, **evenly spaced** and **aligned**.

Contacts: where the resistance is paid

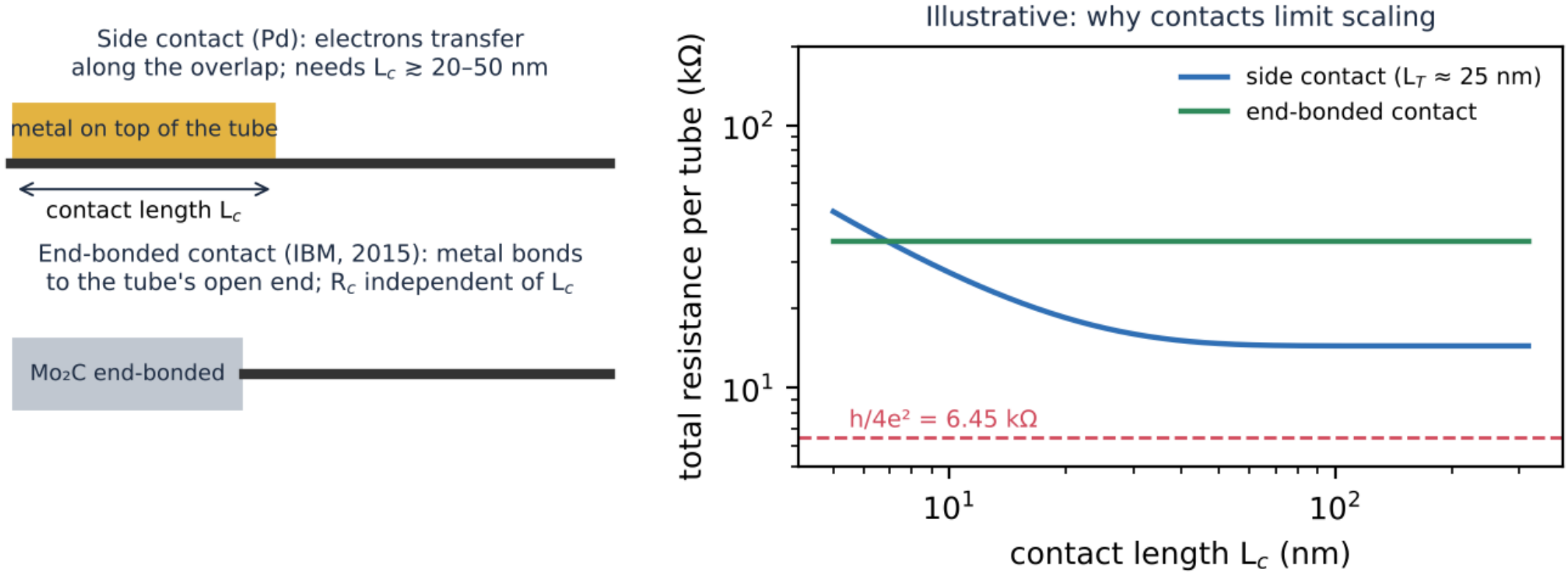
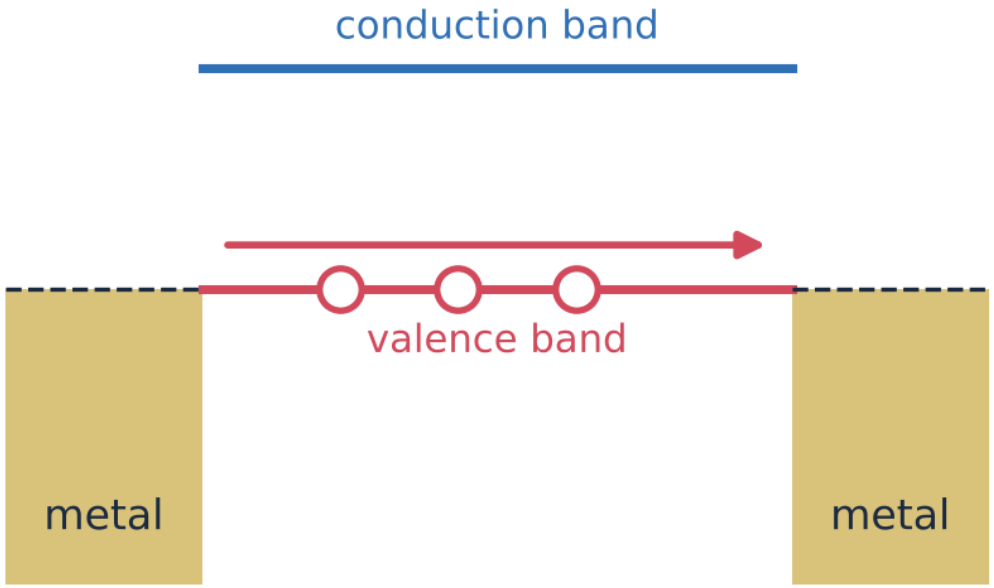


Figure made for this course after Franklin and Chen (2010) and Cao et al. (2015).

A metal laid **on top** of the tube needs 20–50 nm of overlap, too long for a small transistor. Metal **bonded to the tube's end** (IBM, 2015) works at any length.

Choosing the metal chooses n or p

palladium (Pd): lines up with the valence band
holes flow freely: a p-type transistor



scandium (Sc): lines up with the conduction band
electrons flow freely: an n-type transistor

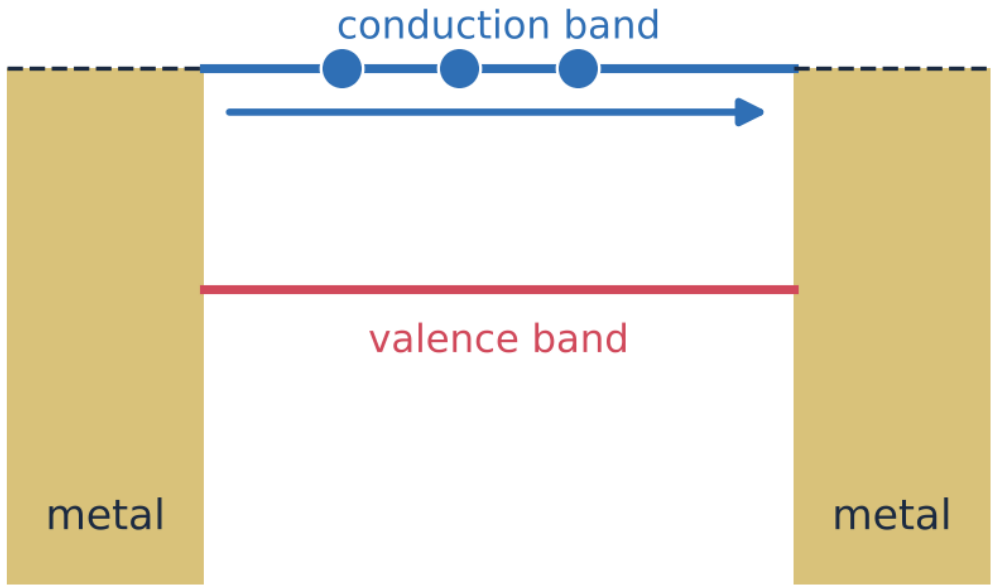


Figure made for this course.

There is no doping as in silicon. **Palladium** lines up with the valence band and makes p-type transistors; **scandium** lines up with the conduction band and makes n-type.

From one tube to a processor

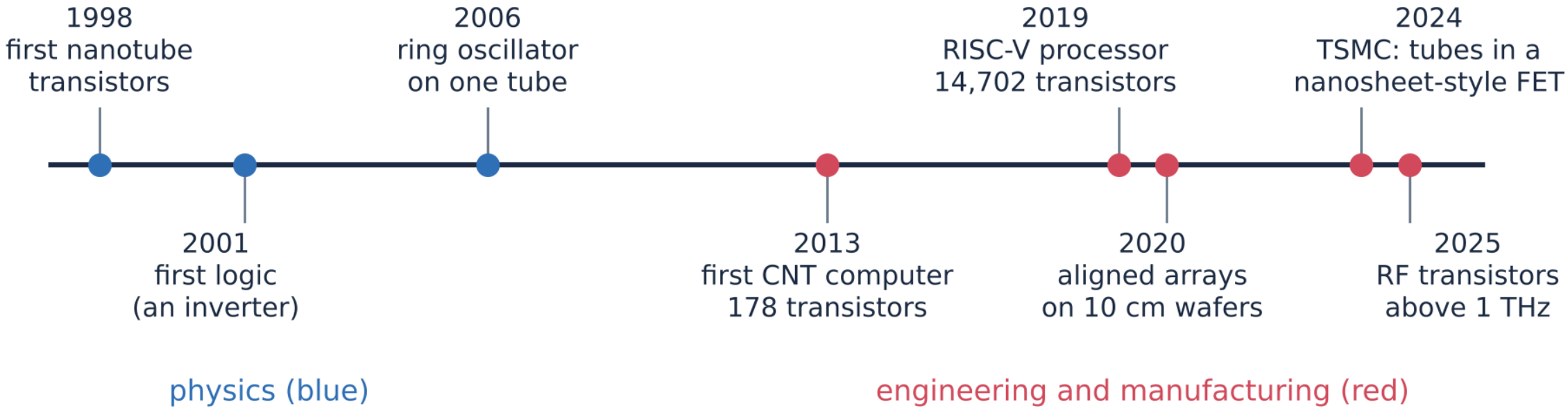


Figure made for this course; events from Document 2, Section 8.

2013 Stanford's first nanotube computer (178 transistors). **2019** MIT's RISC-V processor (14,702). **2020s** wafer-scale aligned arrays; TSMC research devices; RF above 1 THz.

Where nanotube transistors stand in 2026

	Silicon nanosheet (2 nm class)	Nanotube array (best research)
Current per µm	≈ 1–2 mA	1–3 mA reported
n and p	Holes weaker	Symmetric
Largest circuit	≈ 10^{11} transistors	≈ 15,000 transistors
Process temperature	> 1000 °C	< 400 °C: can go on top of silicon

Still needed: **uniform arrays**, **clean gate dielectrics**, **short contacts** and **scale**. First uses: extra layers on silicon chips, RF, flexible electronics and sensors.

Part 1 in six points

1 The tube

Rolled graphene, 1–2 nm; one in three is metallic; gap ≈ 0.8 eV ÷ d.

2 Two ways to switch

Thin the contact barrier, or lower a hill in the channel.

3 Two capacitors

Insulator and quantum capacitance in series.

4 Ballistic

Four lanes, no collisions; 6.45 kΩ per tube at the contacts.

5 The problems

Metallic tubes, contacts, polarity, uniform arrays.

6 Today

Processors of 15,000 transistors; first uses beside silicon.

Credits

Figures

Graphene and rolling (computed atom positions), metal/semiconductor mix, gap, switching, capacitor, lanes, metallic-tube, contact-metal and timeline pictures, device geometries, I–V and contact figures made for KKKC6353 (scripts in figures-source). Historical figures reproduced from T. Ragheb's Rice University ELEC 527 slides (2007) in the old notes: Tans et al., Nature 393, 49 (1998); Derycke et al., Nano Lett. 1, 453 (2001); Chen et al., Science 311, 1735 (2006). Animation screenshots from faizverse.github.io/nanoelectronic-devices.

Sources

Saito, Dresselhaus and Dresselhaus (1998); Martel et al. (1998); Heinze et al. (2002); Javey et al. (2003); Rahman et al., IEEE TED 50, 1853 (2003); Franklin and Chen (2010); Shulaker et al., Nature 501, 526 (2013); Cao et al. (2015, 2017); Hills et al., Nature 572, 595 (2019); Liu et al., Science 368, 850 (2020); TSMC Research (2024); Zhou, Pan et al. (2025). Full list in L4 Document 2.